

U-Net Chest X-ray Lung Segmentation Project Guide

1. Introduction

This project focuses on semantic segmentation of chest X-ray images using the U-Net architecture. The goal is to accurately segment lung regions using a combination of data preprocessing, model training, post-processing, and deployment via FastAPI.

2. Project Structure

The project consists of the following main components:

- `model.py`: Defines the U-Net architecture.
- `preprocess_image.py`: Prepares the input image (resizing, normalization, tensor conversion).
- `postprocess.py`: Cleans the output mask using morphological operations.
- `main.py`: FastAPI server exposing an endpoint for prediction.
- `utils.py`: Contains helper functions.
- `train.py`: (Optional) Script for training and saving the model.

3. Inference Workflow (FastAPI)

1. Upload image via POST `/predict`
2. Image is converted to RGB and preprocessed with albumentations
3. Forward pass through the model
4. Binary threshold applied (output > 0.5)
5. Mask cleaned using `postprocess`
6. Final mask returned as PNG image

4. Preprocessing Function

Uses albumentations to resize image to 256x256, normalize pixel values with mean/std 0.5, and convert to PyTorch tensor.

5. Postprocessing Function

Uses `skimage.morphology.remove_small_regions` to remove small regions (area < 300), and applies morphological closing with OpenCV to smooth the mask and fill holes.

6. Loss Functions Used

A combination of Binary Cross Entropy (BCE) and Dice Loss was used to handle both pixel-level accuracy and structural similarity, especially helpful for class imbalance.

7. Model Saving/Loading

Model saved using `torch.save(model.state_dict(), 'model.pth')`

Model loaded with `model.load_state_dict(torch.load('model.pth'))`

8. Deployment with FastAPI

Run the FastAPI app (e.g., in Google Colab) using `uvicorn main:app --reload`.

Use ngrok to expose the endpoint publicly if needed.